

L30: Sensitivity Analysis

Simulation Without a Black Box

Outline

One-at-a-Time Analysis

Tornado Diagram

Morris Screening

μ -star vs sigma Plot

Input Uncertainty Propagation

Key Takeaways

OAT: One-at-a-Time Sensitivity

OAT Procedure

Fix all parameters at their nominal values. Vary one parameter at a time by $\pm 20\%$ (or $\pm 1\sigma$). Record the change in the response KPI.

Example — clinic W_q at nominal $\rho = 0.85$:

Parameter	-20%	Nominal	+20%
λ	11.3	18.4	31.7
μ	31.7	18.4	11.3
c	18.4	18.4	8.1
CV_{svc}	14.2	18.4	23.1

Swing = $\max W_q - \min W_q$ for each parameter.

```
import numpy as np

params_nominal = {
    "lam": 14.2, "mu": 1/12.7,
    "c": 1, "cv": 0.61
}
deltas = [-0.20, 0.0, +0.20]
results = {}

for param, val in params_nominal.items():
    row = []
    for d in deltas:
        p = params_nominal.copy()
        p[param] = val * (1 + d)
        row.append(
            run_clinic(p, reps=30)
        )
    results[param] = row

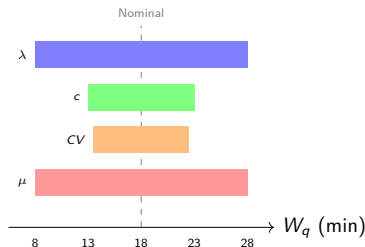
# Swing for each parameter
swings = {k: max(v)-min(v)
          for k,v in results.items()}
```

Tornado Diagram: Ranking Parameters by Swing

Construction:

1. Compute swing = $W_q^{(+20\%)} - W_q^{(-20\%)}$ for each parameter
2. Sort parameters by |swing| descending
3. Draw horizontal bars centered at the nominal W_q

Parameter	Swing (min)
λ	20.4
μ	20.4
CV_{svc}	8.9
c	10.3



OAT Limitation

OAT does not detect interactions. If λ and μ move together (as in rush-hour), the true swing is not captured by varying each alone.

Morris Screening: Elementary Effects

Elementary Effect

For parameter x_i perturbed by Δ :

$$EE_i = \frac{Y(\mathbf{x} + \Delta \mathbf{e}_i) - Y(\mathbf{x})}{\Delta}$$

Compute r elementary effects per parameter at different base points $\mathbf{x}^{(j)}$.

Summary statistics:

$$\mu_i^* = \frac{1}{r} \sum_j |EE_i^{(j)}| \quad \sigma_i = \text{std}(EE_i^{(1)}, \dots, EE_i^{(r)})$$

- **High μ_i^* :** parameter has large overall influence
- **High σ_i :** nonlinear behavior or strong interactions

Advantage over OAT: Morris explores the full parameter space ($r \times (k + 1)$ model evaluations instead of $2k + 1$).

```
import numpy as np

def morris_ee(model, x_nom, deltas,
              r=10, seed=42):
    rng = np.random.default_rng(seed)
    k = len(x_nom)
    ees = {i: [] for i in range(k)}

    for _ in range(r):
        # Random base point
        base = rng.uniform(0.2, 0.8, k)
        x0 = x_nom * base / 0.5
        y0 = model(x0)
        for i in range(k):
            x1 = x0.copy()
            x1[i] += deltas[i]
            y1 = model(x1)
            ees[i].append(
                (y1-y0)/deltas[i])

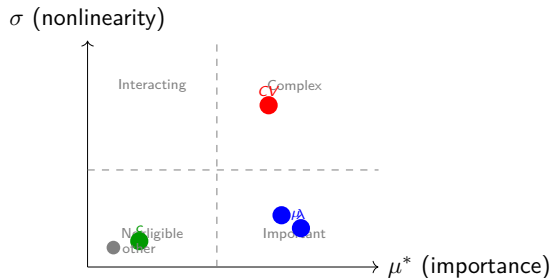
    mu_star = {i: np.mean(np.abs(v))
               for i,v in ees.items()}
    sigma = {i: np.std(v)
             for i,v in ees.items()}
    return mu_star, sigma
```

μ^* vs. σ Plot: Importance and Nonlinearity

Four quadrants:

- Low μ^* , low σ : **negligible** — safely fixed
- High μ^* , low σ : **important, linear** — collect more data on this parameter
- High μ^* , high σ : **important, nonlinear/interacting** — investigate further with full factorial
- Low μ^* , high σ : **interacts but unimportant alone** — watch for interactions with high- μ^* parameters

Rule: Prioritize data collection for parameters in the high- μ^* region — they have the greatest influence on output uncertainty.



Input Uncertainty Propagation

Scenario: We don't know λ exactly — only that it is approximately 14.2 with uncertainty. Suppose $\lambda \sim \mathcal{N}(14.2, 1.5^2)$.

Propagation procedure:

1. Sample M parameter vectors $(\lambda_m, \mu_m, \dots)$ from their marginal distributions
2. Run the simulation model for each parameter vector
3. Summarize the resulting output distribution

```
M = 200
lam_samples = rng.normal(14.2, 1.5, M)
mu_samples = rng.normal(1/12.7, 0.01, M)

Wq_out = [run_clinic(lam=l, mu=m)
           for l,m in
           zip(lam_samples, mu_samples)]

p5 = np.percentile(Wq_out, 5)
p95 = np.percentile(Wq_out, 95)
print(f"Wq: [{p5:.1f}, {p95:.1f}]")
```

Interpretation

The 5th/95th percentile range of W_q quantifies *input uncertainty* — distinct from the statistical uncertainty addressed by CIs.

Uncertainty source	Reduced by
Statistical (sampling)	More replications
Model form	Better data / domain expertise
Input parameter	Better data collection

High μ^* parameters drive input uncertainty. Invest data-collection resources there first.

Key Takeaways

1. **OAT tornado**: quick visual ranking; swing = max — min across the $\pm 20\%$ range. Easy to implement, misses interactions.
2. **Morris μ^*** : average absolute elementary effect over the full parameter space; σ flags nonlinearity.
3. **μ^* vs. σ plot**: four-quadrant diagnostic for importance and interaction structure.
4. **Input uncertainty propagation**: sample parameters from their marginal distributions; output 5th/95th percentile captures model risk.
5. **High $\mu^* \Rightarrow$ collect more data**: sensitivity tells you *where* to spend the field-data budget.
6. **Sensitivity $\neq$ decision importance**: domain knowledge can override statistics (a parameter may be “sensitive” but uncontrollable, or “insensitive” but ethically critical).

Next Lecture

L31 — Ranking and selection: choosing the best system with guaranteed probability of correct selection.